

ALDO ZETINA MUÑOZ

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Experience

- Software Engineer Intern****Banco de México***Fall 2024 – Spring 2026*
- Built an internal fullstack web platform automating institutional email workflows: **Java/Spring** backend exposing **REST** endpoints over a relational store, **JavaScript** frontend, and **Python** services, integrated with the bank's user-directory infrastructure.
 - Reduced pipeline failures by **30 %** by implementing data validation and sanitization in **Python** with retry logic over critical production financial data pipelines.
 - Automated identity management for 500+ users using **PowerShell** and **Python**, integrating **Active Directory** with relational (**SQL**) access-control systems.
 - Implemented audit controls and event logging over financial data flows, improving traceability and observability across production banking systems.
- Fullstack & Multiplatform QA Engineer****Grupo Salinas (Elektra.mx)***Summer 2022 – Spring 2024*
- Implemented core **shopping cart** features in **Java** during the early build phase of Elektra's high-traffic e-commerce mobile app, shipping product code alongside the engineering team.
 - Built a scalable cross-platform testing framework for native and hybrid applications using **Java**, **Selenium**, **Cucumber** and **Maven**, applying object-oriented design and integrating it into CI/CD pipelines with **AWS CodeCommit**.
 - Reduced manual testing time by **40 %** by automating cross-platform suites for **iOS** and **Android** with **Appium**, enabling simultaneous multiplatform release cycles.
 - Validated critical user flows on Elektra's high-traffic app, documenting root causes in **Jira** and preventing cross-release regressions.
- NoSQL Database Engineer****Huawei***Summer 2021 – Spring 2022*
- Built client/server utilities in **C++** over UNIX domain sockets for real-time capture and persistence of network events on distributed Linux systems.
 - Automated network infrastructure monitoring (latency, availability, polling) using **Bash** and **C++**, enabling continuous observability on production Linux environments.
 - Diagnosed L3/L4 incidents by analyzing traffic and logs with **Wireshark** and **tcpdump**, accelerating root-cause resolution in production.

Education

- B.S. in Computer Engineering****UNAM – Faculty of Engineering***Graduation: Summer 2025*
- GPA: 8.5/10.0 (equiv. 3.4/4.0) | Relevant Coursework: Data Structures, Databases, Object-Oriented Programming, Software Engineering, Operating Systems, Distributed Systems.

Technical Experience

- Projects**
- Institutional Email Automation Platform – Banco de México** (2024–2026). Fullstack internal platform: **Spring** backend serving **REST** endpoints over a relational store, web frontend, and integration with the bank's user-directory infrastructure. *Java, Spring Boot, JPA, SQL, JavaScript*
 - Elektra Mobile App – Shopping Cart** (2022–2024). Core shopping-cart functionality implemented in **Java** during the initial build phase of Elektra's e-commerce mobile product. *Java*
 - Cross-Platform Mobile App** (2025–Present). Multiplatform application delivering native-like experiences on iOS and Android from a unified codebase; **REST API** integration, reactive state management, adaptive UI. *Flutter, Dart*

Certifications and Awards

- Advanced SQL** – PROTECO UNAM (Honorable Mention): Complex query design, stored procedures, and database optimization.
- Java Developer** – PROTECO UNAM: OOP, data structures, and application development.

Languages and Technologies

Languages: Java (Advanced), SQL (Advanced), C++ (Advanced), Python (Intermediate), JavaScript, Bash, PowerShell
Backend & Data: Spring / Spring Boot, JPA/Hibernate, JUnit, Maven/Gradle, RESTful APIs, relational databases, stored procedures, query optimization
Tools & Practices: Git, Docker, Linux, CI/CD (AWS CodeCommit), Jira, UML, Active Directory
Testing & Mobile: Selenium, Cucumber, Appium, Flutter/Dart, native & hybrid app testing
Spoken Languages: Spanish (Native), English (Advanced), Japanese (Basic)